The University of Azad Jammu & Kashmir

The University of Azad Jammu and Kashmir

Department of Software Engineering

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Program :BS Software Engineering

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Course Code: SEC-532

Course Instructor: Engr. Ghulam Rabbani Butt

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Lab No: 02

Lab Title: From Real World Concept to Software Object (Library Management System)

Assigned On: October 21, 2026

Due Date: October 28, 2026

Domain: Library Management System

Main functions: Borrowing books, reserving books, managing overdue fines, and handling memberships.

Task 1: List 6–10 Candidate Conceptual Classes

Using the noun-phrase technique and category-list technique, we can identify these conceptual classes.

No.	Candidate Class	Technique	Purpose
1	Book	Noun phrase	Stores book information such as title and ISBN.
2	Member	Noun phrase	Represents a registered library user.
3	Librarian	Noun phrase	Represents the staff member managing library activities.
4	Borrowing	Noun phrase	Records a book issued to a member.
5	Reservation	Noun phrase	Records a member’s request for an unavailable book.
6	Fine	Noun phrase	Represents a penalty for returning a book late.
7	Membership	Category list	Stores membership details and validity.
8	Book Copy	Category list	Represents an individual physical copy of a book.
9	Payment	Category list	Records payments made against fines.

Task 2: Attributes and Methods for Three Important Classes

1. Book

Attributes (State) — describe the book's data.

- bookID
- title
- author
- ISBN
- availabilityStatus

Methods (Behavior) — describe what the book object can do.

- checkAvailability() — checks whether a copy is available.
- updateAvailability() — updates the availability status.
- getBookDetails() — returns the book's information.

2. Member

Attributes (State)

- memberID
- name
- contactNumber
- membershipStatus
- outstandingFine

Methods (Behavior)

- borrowBook() — requests to borrow a book.
- reserveBook() — requests a reservation.
- returnBook() — initiates the return of a borrowed book.
- payFine() — records payment of a fine.

3. Borrowing

Attributes (State)

- borrowingID
- memberID
- bookCopyID
- borrowDate
- dueDate
- returnDate

Methods (Behavior)

- issueBook() — records the issuance of a book.
- calculateOverdueDays() — calculates the number of days past the due date.
- markAsReturned() — records the return of the book.
- checkOverdueStatus() — checks whether the book is overdue.
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Task 3: Abstraction Decisions

In the Library Management System, I focused on the essential details required for borrowing books, reservations, overdue fines, and memberships. I ignored physical details such as a book's color, weight, cover design, and shelf appearance because

they are not necessary for the system's main operations. I also ignored members' personal interests and unrelated personal information to keep the system simple and relevant. This abstraction reduces complexity by representing only the information and behaviors needed to manage library services efficiently.

Task 4: One Invented Software Object

Class: LibraryController

The LibraryController is an invented software object that does not exist as a physical object in a real library. It coordinates interactions between members, books, borrowing records, reservations, and fines.

Possible methods:

- processBorrowRequest() — processes a book-borrowing request.
- processReservation() — manages reservation requests.
- calculateFine() — calculates an overdue fine.
- validateMembership() — checks whether a membership is valid.

Why it is useful: It separates the system's coordination logic from the conceptual classes, making the software easier to organize, maintain, and test.
